



内 蒙 古 大 学
《移动通信》期末考试试卷(A 卷)
(2023 学年 春 季学期)

闭卷 120 分钟

学号_____姓名_____专业_____年级_____ 重修□ 缓考□

总分		题号	一	二	三	四	五	六
阅卷人		评分						

一、简答题(本题共3小题，每小题 10 分，共 30 分)

1. What are the specific advantages of static channel allocation over dynamic channel allocation strategies?
2. What is the difference between the" guard band" and the" guard time", and why are they important in a cellular system?

3. What challenges to be addressed in artificial intelligence and wireless communications for 6G?

二、翻译题(本题共 2 小题,每小题 10 分，共 20 分)

1. A cellular system requires a fairly complex infrastructure. Each Base Station (BS) consists of a Base Transceivers System (BTS) and a BS Controller(BSC). Both tower and antenna are part of the BTS, and all associated electronics are contained in the BTS. The Authentication Center (AUC) unit provides authentication and encryption parameters that verify the user's identity and ensure the confidentiality of each call. The AU protects network operators from different types of frauds and spoofing found in today's cellular world. The Equipment Identity Register(EIR) is a database that contains information about the identity of mobile equipment. Both AU and EIR can be implemented as individual stand-alone units or as a combined AUC/EIR unit. The Home Location Register (HLR) and Visitor Location Register (VLR) are two sets of pointers in databases that support mobility and enable the use of the same number(or mobile phone) over a wide range. The HLR is located at the MSC where the MS is initially registered and is the initial home location for the MSC where the MS is initially registered and is the initial home location for billing and access information.

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四、计算题(第 1 、 2、 3小题各 10 分 , 第 4 小题 20 分 , 共 50 分)

1. A vehicle is moving directly away from the BS with a moving speed of 60 km/h. The BS has a 900 MHz transmitter. If the ratio ρ between specified level and the rms amplitude of the fading envelope is 0.2 and the travel speed of light $c=3\times10^8$ m/s, compute:(a) The fading rate; (b) The fading duration;(c) The level crossing rate.

2. In the next 10 to 20 years, our general mission is to facilitate various applications of machine learning (ML) technologies through intelligent communications. We would like to clarify a couple of concepts between artificial intelligence(AI) and wireless networks, including AI for communication network (AI4Net) and communication network for AI (Net4AI). We demonstrate their correlation and difference at the link level and the network level, respectively. Let us take the link level as an example. Since several years ago, we have been focusing on using AI to improve the intelligence and the transmission performance and reduce the complexity of communication systems as in reference [1], which is regarded as AI4Net. In the future, wireless communications will be an integrated part of AI for performance optimization in addition to data collection and transmission, which is Net4AI. We believe that AI4Net mainly belongs to 5G and 5.5G while Net4AI is the mainstay for the sixth generation(6G) and future wireless networks. For 6G and beyond, AI will be used everywhere in our daily life and communication networks will transmit and collect massive amount data required by ML. Therefore, AI will revolutionize 6G and will also be the fundamental business for 6G. AI in 6G is far more than just over-the-top (OTT) applications. For 6G and beyond, we would put particular focus on Net4AI.

2. It assumed that the base station provides 10 channels, each user keeps a traffic channel busy for an average of 5% time, and an average of 180 requests per hour is generated, compute: (a) the Erlang value; (b) Erlang B blocking probability.

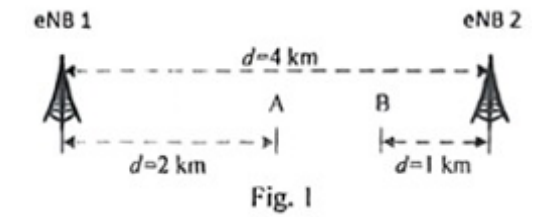
Cell number	Average number of calls/unit time	Cell number	Average number of calls/unit time
1	900	5	1200
2	2000	6	1800
3	2500	7	1000
4	1100		

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3. In a cellular system with seven-cell clusters, the average number of calls at a given time is given as follows:

If the system is assigned 49 traffic channels, how would you distribute the channels if

- Static allocation is used based on traffic load?
- An FCA simple borrowing scheme is used (no traffic load considered)?
- A dynamic channel allocation scheme is used?



4. As shown in Fig.1, eNB 1 and eNB 2 are 4 km apart. The received power of the UE is $P_r = P_t / L(d)$, where P_t is the transmitting power, $L(d)$ is the pathloss in dB and is given by $L(d) = 20\alpha \log_{10}(d) + X$, d is the distance between eNB 1 and eNB 2, α is the path loss coefficient, X is the shadow fading and $X \sim N(0,9)$.

- What is the received power in dBm and signal-to-interference ratio from eNB 1 when UE is located at A under the condition that shadow fading is ignored? (Assuming $P_{t1} = 2W$, $P_{t2} = 2W$, $\alpha = 4$).
- What is the probability that the received power from eNB 2 above -90 dBm and when of UE is located at B? (Assuming $P_{t1} = 2W$, $P_{t2} = 2W$, $\alpha = 4$)
- What is the probability that signal-to-interference ratio from eNB 2 above 3 dB and when of UE is located at B? (Assuming $P_{e1} = 4W$, $P_{t2} = 2W$, $\alpha = 4$)
- How to adjust transmit power of eNB 1 if the handover is occurred at location B? Handover is triggered when $R_2 > R_1 + T$ is satisfied. ($P_{t2} = 2W$, $\alpha = 4$, non-shadow fading, $T = 3dB$)

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